21 A spring is fixed at one end and extended by applying force F to the other end. The spring has extension x and elastic potential energy E_P. The spring constant is k.

The spring obeys Hooke's law.

Which relationship is correct for this spring?

A $E_P \propto F$ **B** $E_P \propto x$ **C** $E_P \propto k$ **D** $E_P \propto x^2$